Pol-ISARSpaceTarget-1.0: Polarimetric ISAR Electromagnetic Simulation Dataset for Space Target Fine Recognition

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Abstract

Space targets, represented by satellites, play important roles in remote sensing and mapping, meteorological monitoring, wireless communication, reconnaissance and surveillance, and are also important objects of space situational awareness. Polarimetric inverse synthetic aperture radar (ISAR) can provide high-resolution target images and polarimetric scattering information that is sensitive to target structure, giving it distinctive advantages for space situational awareness. However, real measured polarimetric ISAR data for space targets are difficult to make publicly available, while fine-grained component-level annotation is difficult and has received limited study. Consequently, publicly available polarimetric ISAR space-target datasets are lacking. To address these challenges and promote research in this field, the authors construct and release Pol-ISARSpaceTarget-1.0, described as the first publicly available electromagnetic-simulation dataset for fine recognition of polarimetric ISAR space targets. It contains polarimetric ISAR image data for six types of space targets—including radar satellites with parabolic antennas, radar satellites with panel antennas, a communication satellite, and an optical satellite—and semantic annotations for nine key component classes. The paper describes the dataset composition, electromagnetic simulation and imaging procedures, and component annotation workflow. Using space-target component recognition as an example, representative deep-learning oriented-object-recognition methods are evaluated to validate the dataset and establish benchmark results. The dataset provides diverse space-target types, complete polarimetric information, and rich component annotations, offering foundational data support for fine-grained classification and recognition of space targets.

Keywords: radar polarimetry; inverse synthetic aperture radar; space target; dataset; target recognition

1. Introduction

Space targets represented by satellites are increasingly important in remote sensing and mapping, meteorological monitoring, wireless communication, reconnaissance and surveillance, and other applications. The number of on-orbit space targets has increased rapidly in recent years, creating an urgent need for timely and accurate space situational awareness. Major technical means for space situational awareness include radar detection, electro-optical detection, infrared detection, and radio detection. Among these, inverse synthetic aperture radar (ISAR) provides long-range, high-resolution, all-time and all-weather monitoring of space targets and has therefore received substantial development.

In 1970, the United States completed ALCOR, the world's first high-power, long-range, wideband imaging radar. To meet deep-space exploration requirements, the United States subsequently developed the Goldstone, Haystack, and HUSIR radars; imaging bandwidth increased progressively from 1 GHz to 8 GHz. Polarization is another important information source in addition to the time, frequency and spatial domains of electromagnetic waves, and has been important in enhanced recognition of artificial targets, land-cover classification, disaster assessment, and other applications. A target's polarimetric response is closely related to its structure and material. Polarimetric radar, represented by polarimetric ISAR, can therefore provide rich information for space-target detection and recognition. For example, Goldstone radar measurements have shown that jointly using dual-polarization information can significantly improve target detection performance. Polarimetric ISAR has consequently become an important sensor for space-target detection and recognition.

Fine-grained recognition of space targets can help determine their health status, payload type, and behavioral status, thereby providing key information for operational-state monitoring and inference of behavioral intent. For example, Lincoln Laboratory in the United States used the ALCOR radar to obtain high-resolution images and conduct state monitoring of space targets including the former Soviet Salyut-1 space station and the U.S. Skylab space station. Solar arrays are a typical space-target component: their on-orbit operating state directly affects power supply to the payload and, to some extent, the lifetime of the space target. Antennas are key components of radar remote-sensing satellites, and their on-orbit attitude is important for analyzing the target's operating state and potential operational intent.

Deep-learning detection and recognition methods have developed rapidly. Compared with traditional algorithms, they do not require manually designed features and can be trained end-to-end, and they have gradually become mainstream for detection and recognition. Dataset construction is important for the development of deep-learning applications and interdisciplinary research. However, because real measured polarimetric ISAR space-target data are difficult to make public for confidentiality and other reasons, research teams in China generally use electromagnetic computational simulation to construct polarimetric ISAR space-target datasets. Existing studies have mainly focused on classification and recognition of the overall target category.

The paper reviews prior simulation datasets: one study constructed a 1-GHz-bandwidth single-polarization satellite simulation dataset for target-level classification and recognition; two studies constructed 2-GHz-bandwidth single-polarization satellite simulation datasets for attitude estimation and target-level classification/recognition; another constructed a 1.5-GHz-bandwidth fully polarimetric satellite dataset for abnormal-attitude estimation; another constructed a 2-GHz-bandwidth fully polarimetric satellite dataset for classification under unknown target deformation. At higher, terahertz frequencies, point-target simulation has also been used to construct ISAR space-target datasets and study imaging characteristics.

The authors note that these electromagnetic simulation datasets had not been publicly released. Researchers therefore still needed to collect or build satellite models themselves and perform electromagnetic calculations and imaging processing, which is labor-intensive and unfavorable to technical development and communication. Existing space-target ISAR simulation datasets also primarily concern the overall target. In addition, radar images are more difficult to visually interpret than optical images, making fine-grained component annotation difficult. To address these conditions and challenges, the authors construct and release Pol-ISARSpaceTarget-1.0, containing six space-target types and nine key component classes, and provide baseline results for component recognition using representative deep-learning object-recognition methods.

2. Dataset Construction

2.1 Dataset Composition

Pol-ISARSpaceTarget-1.0 contains 1,638 frames of polarimetric ISAR images for six space-target types and 16,577 annotated instances belonging to nine key component classes. The polarization channels are HH, HV, VH and VV. To facilitate full use of amplitude and phase information, the dataset provides complex-valued data.

The dataset is organized into four top-level directories: Annotations, ImageSets, TIFFImages and Tools. Annotations contains 1,638 XML annotation files for the six satellite targets. ImageSets contains 13,104 DAT-format complex-valued data files—covering the four polarization channels for the 1,638 frames—and 1,638 TXT annotation files. Data from all viewing angles are split into training, validation and test sets in an 8:1:1 ratio. TIFFImages contains 13,104 TIFF amplitude images and 1,638 TIFF Pauli pseudo-color images. Tools contains Python and MATLAB functions for reading DAT-format complex-valued data.

Satellite	Type	Length (m)	Width (m)	Height (m)
HISEA-1	Remote-sensing satellite	4.0	1.1	0.4
RADARSAT-2	Remote-sensing satellite	15.0	1.5	3.2
StarLink	Communication satellite	4.0	1.8	0.9
WorldView-4	Optical satellite	8.0	8.0	5.3
Capella	Remote-sensing satellite	4.6	3.5	4.0
QPS-SAR	Remote-sensing satellite	3.6	3.6	2.0

Table 1. Satellite categories and dimensions.

2.2 Electromagnetic Characteristics Calculation and Imaging Processing

Because real measured polarimetric ISAR space-target data are difficult to obtain, the research team performs electromagnetic simulation using detailed CAD models of representative space targets: HISEA-1, RADARSAT-2, StarLink, Capella, QPS-SAR (Q-shu Pioneers of Space-Synthetic Aperture Radar), and WorldView-4. The selected satellites have different geometric configurations and functions and cover radar/remote-sensing, communication, and optical satellite types. Their dimensions are listed in Table 1 and their detailed CAD models are shown in Figure 2.

First, full-polarimetric electromagnetic characteristics of the six satellites are calculated in the Ku band using the physical-optics method. The electromagnetic calculation uses a 16-GHz center frequency, 2-GHz bandwidth, 0.01-GHz frequency step, and 0.025° azimuth step. After obtaining the target electromagnetic characteristics, imaging is centered initially at 0° azimuth, with a 1° step between successive imaging-center azimuths. At three elevation/depression angles—40°, 45° and 50°—a polar-format algorithm is applied to the full-polarimetric RCS data over a 2-GHz bandwidth and a 5° coherent integration angle to generate polarimetric ISAR images, forming the ImageSet image dataset.

The ImageSet images are then processed by frequency-domain interpolation and image cropping to produce 256 × 256-pixel, 64-bit complex-valued data. Each complex value contains a 32-bit floating-point real part and a 32-bit floating-point imaginary part, and the four polarization channels are stored separately in DAT format. For visualization and compatibility with mainstream deep-learning workflows, the authors provide TIFF images in two subsets: AmplitudeImages and PauliImages.

Amplitude is a primary radar-data feature and reflects the strength of backscattering from different objects. The 64-bit complex data are therefore converted into 32-bit floating-point amplitude data; after linear grayscale stretching for visualization and data-volume considerations, they are stored as 8-bit integer TIFF data, forming AmplitudeImages. After Pauli decomposition of the full-polarimetric complex data, Pauli pseudo-color images are constructed and stored as 8-bit integer TIFF data, forming PauliImages. In the Pauli pseudo-color representation, the red, blue and green components correspond to the polarimetric features |S_HH − S_VV|, |S_HH + S_VV| and |S_HV|, respectively.

Parameter	Value
Center frequency	16 GHz
Bandwidth	2 GHz
Azimuth range	–5° to 95°
Elevation/depression-angle range	40°, 45°, 50°
Polarization	HH, HV, VH, VV
Electromagnetic calculation method	Physical optics (PO)

Table 2. Electromagnetic simulation parameters.

2.3 Space-Target Component Annotation

Component-level recognition of space targets can help infer payload type and operational status. However, no publicly released component-annotation dataset for polarimetric ISAR space targets was available. Unlike optical images, radar images are more complex: scattering characteristics of different components are mutually coupled and are difficult to interpret directly by visual inspection, making fine component annotation challenging.

To address this challenge, the research team proposes a three-dimensional-point-cloud-assisted space-target component annotation method. First, a 3-D point cloud is generated from the 3-D model of the space target. The point cloud is then annotated by component. Using the imaging geometric relationship, each component's 3-D point cloud is projected onto the two-dimensional imaging plane. Because components have orientation, an oriented bounding box is generated from the enclosing geometry of each component's point cloud to more accurately identify its spatial position, producing a coarse component annotation. Finally, the annotation parameters are manually checked and corrected one by one against the 3-D space-target model under different imaging views, yielding the final oriented-box semantic annotations.

The dataset uses a two-level component taxonomy. The first-level categories are Body, Solar Array, Antenna, Fitting and Optical Lens. At the second level, categories are further divided according to physical function and geometric shape. Considering the characteristics of polarimetric ISAR images, nine key component classes are emphasized: Cuboid Body, Polygon Body, Plate Body, Solar Array, Parabolic Antenna, Panel Antenna, Plate Fitting, Rod Fitting and Optical Lens. Across the six space targets and their different viewing angles, 16,577 oriented bounding-box component instances are obtained.

For oriented-box parameters, the upper-left corner of the polarimetric ISAR image is the coordinate origin (0,0), the horizontal image direction is the x-axis, and the vertical direction is the y-axis. The dataset provides both DOTA and PASCAL VOC component annotation formats. In DOTA, the TXT file records the x/y coordinates of the four oriented-box

corners, the component class, and recognition difficulty; difficulty 0 denotes a non-difficult sample and 1 denotes a difficult sample. For PASCAL VOC, the 'le90' representation defines w as the long side, h as the short side, and θ as the angle between the long side and the x-axis. $\theta \in [-\pi/2, \pi/2)$, with clockwise angles positive and counterclockwise angles negative. The oriented box is recorded as (c_x, c_y, h, w, angle) in XML, together with the filename, satellite name, imaging-center azimuth, imaging-center elevation/depression angle, and oriented-box parameters.

2.4 Dataset Analysis

2.4.1 Dataset Distribution

The distribution analysis in Figure 5 shows that the number of fitting components is relatively large in version 1.0, whereas optical-lens components are relatively few. Component-size distributions are also diverse. Body components span approximately 2 × 2 to 65 × 65 pixels; solar-array components are concentrated mainly in 20 × 20 to 40 × 40 pixels; antenna components range from 5 × 5 to 160 × 160 pixels; fitting components are small, with maximum area not exceeding 30 × 30 pixels; optical-lens components are concentrated mainly in 40 × 40 to 45 × 45 pixels. Components with area greater than 100 × 100 pixels are mainly parabolic antennas and solar arrays.

2.4.2 Typical-View Examples

In aspect-ratio and angle distributions, plate-type components have relatively uniform length-to-width distributions. Panel antennas, optical lenses and rod-like fittings have length-to-width ratios mainly below 0.5 and therefore exhibit elongated shapes, whereas parabolic antennas have ratios mainly above 0.5. Component orientations are diverse, but because of imaging projection, the angle distributions are not balanced among component types. Figure 6 presents Pauli pseudo-color images for representative views of the six satellites, with red, blue and green representing |S_HH − S_VV|, |S_HH + S_VV| and |S_HV|. Figure 7 shows oriented-box component annotations, with different colors denoting different component types.

3. Space-Target Component Recognition Experiments

3.1 Recognition Models

Deep-learning object-recognition methods generally contain a backbone network, a neck network and a detection/recognition head. The backbone extracts shallow and multi-scale deep features; the neck fuses multi-scale deep features; and the head estimates bounding-box position, geometry and class parameters from the fused features. Depending on the bounding-box type, methods can be categorized into horizontal/regular-box detection and oriented-box detection. Because space-target components are closely spaced and have strong coupling relationships, oriented-box models are used to accurately localize components.

Six representative deep-learning oriented-object-recognition methods are compared to establish a benchmark: O-FCOS (Fully Convolutional One-stage Object Detection for Oriented Object), PSC (Phase-Shifting Coder), O-RTMDet (Real-Time Models for Object Detection Method for Oriented Object), O-ATSS (Oriented Adaptive Training Sample Selection), S2A-Net (Single-Shot Alignment Network), and O-FRCNN (Faster Region-Based Convolutional Neural Networks Method for Oriented Object).

3.2 Experimental Setup and Evaluation Metrics

The comparison experiments use a server equipped with an Intel Xeon 8336C CPU and an NVIDIA A800 GPU. The recognition algorithms are trained and tested using the mmrotate deep-learning framework with the AdamW optimizer. The base learning rate is 0.001 and weight decay is 0.05. Training initially uses a constant learning rate; at training step 50, cosine annealing is applied with a half-cycle of 50 steps, so that the learning rate at the end of training is 0.05 times its initial value. Batch size is 64, training is 100 steps, and the backbone network is initialized with pretrained weights. The validation set is used during training, and the best-performing weights are saved for testing.

The selected oriented-object methods normally take three-channel RGB images as network input, and their input data and network weights are real-valued. For the complex-valued polarimetric ISAR data, the authors first form the polarimetric scattering vector V = [S_HH, S_VH, S_HV, S_VV]^T. They then construct a real-valued eight-channel input by interleaving the real and imaginary parts:

[Re(S_HH), Im(S_HH), Re(S_VH), Im(S_VH), Re(S_HV), Im(S_HV), Re(S_VV), Im(S_VV)]^T.

Four common target-detection/recognition metrics are used: Recall, Precision, F1 score and mean Average Precision (mAP). The intersection over union (IoU) between a predicted oriented box and a ground-truth oriented box is used with a predefined threshold to determine true positives (TP) and false positives (FP). Normally, a prediction with IoU > 0.5 is treated as a TP; otherwise it is an FP. A false negative (FN) is an annotated oriented box that is not successfully matched.

Precision and recall reflect false-alarm and missed-detection behavior; F1 is their harmonic mean. Because deep-learning methods can jointly perform detection and recognition end-to-end, mAP is used to jointly characterize localization and class-recognition performance; the paper uses mAP50, i.e., mean average precision at IoU = 0.5.

The paper gives the following definitions: Recall = TP/(TP+FN); Precision = TP/(TP+FP); F1 = 2·Precision·Recall/(Precision+Recall). For mAP50, the paper expresses the metric as the average of the precision–recall integral over the N recognized target classes, with N = 9 in this study.

3.3 Comparative Analysis of Experimental Results

To evaluate viewpoint generalization, four subsets are constructed according to imaging elevation/depression angle. Each subset contains all imaging azimuths for the six satellites at one angle and is denoted λ40, λ45, λ50, or λall, where λall = λ40 ∪ λ45 ∪ λ50. Each subset is divided into training, validation and test data in an 8:1:1 ratio. For example, λ40_train denotes the training subset at a 40° imaging angle. All six methods are trained using each of the four subsets and tested separately on λ40, λ45 and λ50, yielding 12 training/testing conditions per method.

Across the 12 training/testing conditions, O-RTMDet has the best mean performance on all four metrics: mean Recall 69.92%, mean Precision 79.11%, mean F1 72.82%, and mean mAP50 68.28%. Except for S2A-Net, all methods have mAP50 above 60% on average. When testing at an imaging angle not represented in training, performance drops substantially. For example, training on the 40° subset and testing on the 50° subset causes O-RTMDet's mAP50 to decrease by 25.60% and its F1 score to decrease by 26.01%. Among the six methods, O-FCOS shows the largest decline under this cross-view condition: mAP50 decreases by 35.50% and F1 by 35.76%. The authors therefore conclude that the dataset provides a useful data basis for component recognition, but the viewpoint generalization of existing deep-learning component-recognition methods still needs improvement.

Table 3. Recall results under different training/testing conditions (%).

Training → Test	O-FRCNN	O-ATSS	S2A-Net	O-FCOS	PSC	O-RTMDet
λ40 train → λ40 test	78.50	78.00	67.56	80.92	71.47	77.42
λ40 train → λ45 test	62.51	65.74	45.97	62.06	58.87	65.02
λ40 train → λ50 test	50.63	46.48	43.02	46.81	47.29	51.83
λ45 train → λ40 test	69.57	70.24	50.66	71.03	70.88	70.98
λ45 train → λ45 test	74.00	73.48	60.06	76.60	71.06	73.17
λ45 train → λ50 test	60.03	56.32	46.50	62.00	56.39	61.82
λ50 train → λ40 test	60.87	60.89	30.10	59.81	54.44	56.09
λ50 train → λ45 test	68.34	66.48	43.76	65.93	68.60	66.17
λ50 train → λ50 test	72.16	71.80	48.87	71.62	66.61	69.26
λall train → λ40 test	82.53	79.37	69.62	80.27	73.23	84.49
λall train → λ45 test	79.17	75.07	63.02	78.24	79.22	82.34
λall train → λ50 test	73.56	71.34	63.64	76.10	76.23	80.40
Mean	69.32	67.93	52.73	69.28	66.19	69.92

Table 4. Precision results under different training/testing conditions (%).

Training → Test	O-FRCNN	O-ATSS	S2A-Net	O-FCOS	PSC	O-RTMDet
λ40 train → λ40 test	76.19	86.85	68.36	84.81	72.30	86.53
λ40 train → λ45 test	52.07	70.43	50.57	64.09	57.45	73.38
λ40 train → λ50 test	40.86	48.15	49.92	46.87	41.39	59.99
λ45 train → λ40 test	64.74	81.44	63.86	72.77	76.11	78.44
λ45 train → λ45 test	70.87	79.46	64.24	79.08	73.59	80.71
λ45 train → λ50 test	56.93	60.04	54.16	64.68	57.34	69.53
λ50 train → λ40 test	52.60	66.22	41.20	63.32	56.06	70.51
λ50 train → λ45 test	61.38	70.70	53.17	67.43	68.29	75.74
λ50 train → λ50 test	70.11	79.14	57.79	75.02	67.28	81.58
λall train → λ40 test	84.42	88.38	72.49	86.76	77.64	93.42
λall train → λ45 test	81.67	83.52	66.97	83.19	83.74	90.19
λall train → λ50 test	75.82	79.19	65.82	81.02	77.95	89.26
Mean	65.64	74.46	59.05	72.42	67.43	79.11

Table 5. F1 results under different training/testing conditions (%).

Training → Test	O-FRCNN	O-ATSS	S2A-Net	O-FCOS	PSC	O-RTMDet
λ40 train → λ40 test	76.98	81.08	67.38	82.37	71.69	80.47
λ40 train → λ45 test	56.34	67.36	47.95	62.81	57.61	67.71
λ40 train → λ50 test	44.84	47.04	45.88	46.61	43.67	54.46
λ45 train → λ40 test	66.42	74.37	54.99	71.59	72.81	72.80
λ45 train → λ45 test	71.58	75.68	61.34	77.50	71.75	75.18
λ45 train → λ50 test	58.02	57.60	49.37	63.10	56.31	64.05
λ50 train → λ40 test	55.23	62.88	34.32	60.58	53.79	60.98
λ50 train → λ45 test	64.00	68.00	47.43	66.47	67.80	69.32
λ50 train → λ50 test	70.42	74.60	52.43	72.95	66.40	73.04
λall train → λ40 test	82.78	82.56	70.80	82.83	74.79	87.28
λall train → λ45 test	79.63	78.07	64.65	80.13	80.98	85.01
λall train → λ50 test	74.04	74.27	64.46	77.96	76.73	83.50
Mean	66.69	70.29	55.08	70.41	66.19	72.82

Table 6. mAP50 results under different training/testing conditions (%).

Training → Test	O-FRCNN	O-ATSS	S2A-Net	O-FCOS	PSC	O-RTMDet
λ40 train → λ40 test	76.00	77.80	65.00	80.40	66.10	75.10
λ40 train → λ45 test	58.10	62.10	44.50	59.30	52.90	62.10
λ40 train → λ50 test	44.10	44.30	39.80	44.90	44.50	49.50
λ45 train → λ40 test	68.10	68.50	47.00	67.00	67.50	68.80
λ45 train → λ45 test	70.60	70.90	55.20	73.80	65.60	73.50
λ45 train → λ50 test	57.60	53.70	44.30	59.20	50.90	59.40
λ50 train → λ40 test	54.60	58.60	27.00	55.90	51.60	53.90
λ50 train → λ45 test	63.90	62.00	38.60	62.40	62.50	64.50
λ50 train → λ50 test	71.00	69.40	40.80	69.10	62.90	69.50
λall train → λ40 test	81.50	78.70	66.90	78.30	70.10	84.60
λall train → λ45 test	75.10	73.40	59.70	75.80	75.40	80.30
λall train → λ50 test	71.00	70.90	60.10	72.70	72.50	78.10
Mean	65.97	65.86	49.08	66.57	61.88	68.28

4. Conclusion

The authors construct Pol-ISARSpaceTarget-1.0 in response to the lack of publicly available datasets for polarimetric ISAR space-target recognition. The dataset contains fully polarimetric ISAR image data and fine component annotations for six representative space targets and nine key component classes, providing basic data support for fine-grained space-target classification and recognition. Future work will expand the number of space-target types and component annotation categories. The research team also plans to build physical space-target models and conduct anechoic-chamber and field measurement experiments to further enrich and improve the dataset.

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Executive Summary: Key Points, Methods, Techniques, and Results

Research problem and motivation

The paper addresses a practical data bottleneck in polarimetric ISAR space-target recognition: real measured data are difficult to make public, and component-level semantic annotation is difficult because radar scattering is complex and components are strongly coupled. Existing simulation datasets cited by the authors were not publicly released and were mainly oriented toward whole-target recognition. Pol-ISARSpaceTarget-1.0 is intended to make a reusable, fine-grained polarimetric ISAR resource available to researchers.

What the dataset contains

The dataset contains 1,638 polarimetric ISAR frames for six satellite targets and 16,577 component annotation instances covering nine component classes. It provides four polarization channels (HH, HV, VH, VV) in complex-valued form, 13,104 DAT complex files, 1,638 TXT annotations, 13,104 TIFF amplitude images, 1,638 TIFF Pauli pseudo-color images, and Python/MATLAB readers. The standard split across all views is 8:1:1 for train/validation/test.

Targets represented

The six detailed CAD models are HISEA-1, RADARSAT-2, StarLink, WorldView-4, Capella and QPS-SAR. Their listed dimensions span substantially different configurations: lengths 3.6–15.0 m, widths 1.1–8.0 m, and heights 0.4–5.3 m. The types cover remote-sensing/radar satellites, a communication satellite, and an optical satellite.

Electromagnetic simulation and imaging pipeline

The electromagnetic simulation uses the physical-optics method in the Ku band, with center frequency 16 GHz, bandwidth 2 GHz, frequency step 0.01 GHz, azimuth step 0.025°, azimuth range −5° to 95°, and elevation/depression angles 40°, 45°, and 50°. Full-polarimetric RCS is imaged using a polar-format algorithm with a 2-GHz bandwidth and 5° coherent integration angle. The resulting images are frequency-domain interpolated and cropped to 256 × 256 complex-valued pixels. Four polarization channels are retained separately. For conventional image pipelines, the data are also rendered as amplitude TIFFs and Pauli pseudo-color TIFFs.

Polarimetric representation

Pauli pseudo-color uses red = |S_HH − S_VV|, blue = |S_HH + S_VV|, and green = |S_HV|. For deep-learning experiments, the complex scattering vector V = [S_HH, S_VH, S_HV, S_VV]^T is converted to an eight-channel real-valued tensor by interleaving real and imaginary components. This preserves both amplitude/phase information and compatibility with real-valued deep-learning architectures.

Component annotation technique

A three-dimensional-point-cloud-assisted workflow is used: generate a 3-D point cloud from the CAD model → assign component labels in 3-D → project component points to the 2-D ISAR image plane using imaging geometry → generate oriented bounding boxes from the projected component envelope → manually inspect and correct the annotations against the 3-D target model for each view. Both DOTA and PASCAL VOC annotation formats are provided.

Recognition models and experimental protocol

Six oriented-object detectors are benchmarked: O-FCOS, PSC, O-RTMDet, O-ATSS, S2A-Net and O-FRCNN. Training uses mmrotate, AdamW, learning rate 0.001, weight decay 0.05, batch size 64, 100 training steps, pretrained backbones, and cosine learning-rate annealing beginning at step 50. The study deliberately evaluates viewpoint generalization by training at λ40, λ45, λ50, or λall and testing at λ40, λ45, or λ50, producing 12 train/test conditions per method.

Evaluation metrics

The reported metrics are Recall, Precision, F1 and mAP50. A predicted oriented box is a true positive when IoU > 0.5; otherwise it is counted as a false positive, while unmatched ground-truth boxes are false negatives. mAP50 is the mean average precision at IoU = 0.5 across the nine component classes.

Main quantitative result

O-RTMDet has the best mean result across the 12 conditions on all four metrics: Recall 69.92%, Precision 79.11%, F1 72.82%, and mAP50 68.28%. This establishes the paper's principal benchmark result for the dataset.

Viewpoint-generalization result

The central limitation is cross-view generalization. Training at 40° and testing at 50° causes O-RTMDet mAP50 to fall by 25.60% and F1 by 26.01%. O-FCOS suffers the largest decline among the six methods in that condition, with mAP50 down 35.50% and F1 down 35.76%. The authors therefore regard viewpoint generalization as an unresolved research problem rather than claiming that existing detectors are robust to unseen viewing angles.

Overall contribution and future work

The paper's contribution is primarily a public, fine-grained polarimetric ISAR simulation resource plus a standardized oriented-component-recognition benchmark. The authors plan to expand the number of target types and annotation categories, and to build physical target models for anechoic-chamber and field measurement experiments so that the dataset can be further enriched with measured data.

Important interpretation / limitation

The dataset is electromagnetic-simulation based, not a collection of real measured on-orbit observations. Therefore, the reported detector results establish a benchmark on this simulated dataset and should not be interpreted by themselves as a validation of real-world operational performance. The paper itself identifies the need for future physical-model, anechoic-chamber and field measurements.

Source-grounded takeaway

The strongest source-supported conclusion is twofold: Pol-ISARSpaceTarget-1.0 materially improves the availability of fine-grained polarimetric ISAR training/evaluation data, while the benchmark experiments show that current oriented-object detectors remain sensitive to changes in imaging viewpoint. The dataset is therefore both a resource and a diagnostic benchmark for future work on polarimetric representation learning, oriented detection, and viewpoint-robust space-target component recognition.